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Research Article

Scalable Enzymatic Synthesis of Free-Reducing Human Milk Oligosaccharides via a β-Galactosidase-Resistant Tagging Strategy

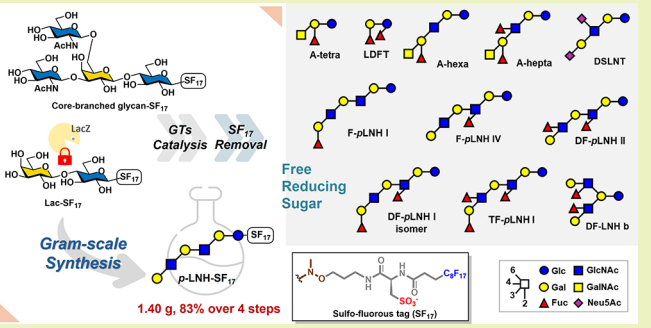
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Cite This: ACS Sustainable Chem. Eng. 2026, 14, 1067−1076

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ABSTRACT: Free-reducing end glycans are essential tools in glycan−protein interaction studies, structural biology, and functional glycomics, owing to their native conformational flexibility and unmodified binding epitopes. Human milk oligosaccharides (HMOs) are bioactive glycans with growing applications in nutrition and therapeutics. While bacterial glycosyltransferases (GTs) are widely used in HMO synthesis due to their ease of recombinant expression in *Escherichia coli*, preparative-scale production requires sufficient amounts of active enzymes, which are often obtained through labor-intensive purification. Using crude cell lysates provides a practical alternative that avoids protein loss during purification, particularly for labile or membrane-bound GTs. However, crude lysates typically contain endogenous hydrolases that can degrade substrates and compromise synthetic efficiency. Here we present a sulfo-fluorous affinity tag (SF_{17}) that enables β-galactosidase-resistant glycan synthesis directly in crude lysates. SF_{17}-tagged glycans show high stability, enhanced regioselectivity, and are readily purified via fluorous solid-phase extraction. This strategy facilitated preparative-scale access to diverse HMOs, including *para*-lacto-*N*-hexaose (*p*LNH), A-antigen derivatives, 2′-fucosyllactose (2′-FL), lactodifucotetraose (LDFT), disialyllacto-*N*-tetraose (DSLNT), and difucosyllacto-*N*-hexaose b (DF-LNH b). The SF_{17} tag is easily removed under mild acidic conditions, yielding pure free-reducing glycans suitable for downstream applications. This work provides a practical and scalable platform that addresses major enzymatic and purification bottlenecks in glycan synthesis.



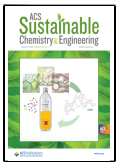
KEYWORDS: *glycosyltransferases, carbohydrate-active enzymes, human milk oligosaccharides, enzymatic carbohydrate synthesis, free reducing end sugars*

INTRODUCTION

Glycan−protein interactions play a crucial role in various biological processes, including immune response, pathogen recognition, and cell signaling.[1,2] The study of these interactions often relies on glycan arrays, in which glycans are immobilized on solid surfaces through linkers.[3−5] While linker-assisted glycoconjugates facilitate high-throughput screening, many biologically relevant glycans exist in their free-reducing form, which better reflects their intrinsic properties. The free-reducing end glycan is critical for preserving native conformational flexibility, binding affinity, and enzymatic reactivity.[6,7] Moreover, crystallographic studies of glycan-protein complexes often require free-reducing glycans to obtain biologically relevant structures, as linkers can interfere with binding or crystallization.[8,9] Certain glycan-binding proteins even recognize the reducing end, further emphasizing the need for native, unmodified glycan structures.[10]

Among such glycans, human milk oligosaccharides (HMOs) represent a prominent class of naturally occurring free-reducing end glycans with well-established roles in infant nutrition, immune regulation, and gut microbiota development.[11−13] Their biological functions and structural diversity make them highly desirable synthetic targets for functional and therapeutic studies.[14,15] Several strategies have been developed for the synthesis of free-reducing HMOs, including microbial fermentation,[16−20] chemical synthesis,[21−24] and enzymatic approaches.[25−28] Microbial fermentation using engineered *Escherichia coli* or *Lactococcus lactis* enables large-scale HMO production,[29] but often yields complex mixtures that require extensive downstream purification. Chemical synthesis provides precise control over glycan structure, but is labor-intensive and not easily scalable. Enzymatic synthesis, which utilizes glycosyltransferases (GTs) and glycosidases, offers superior regioselectivity and efficiency. However, the major

Received: October 2, 2025
Revised: December 25, 2025
Accepted: December 26, 2025
Published: January 2, 2026



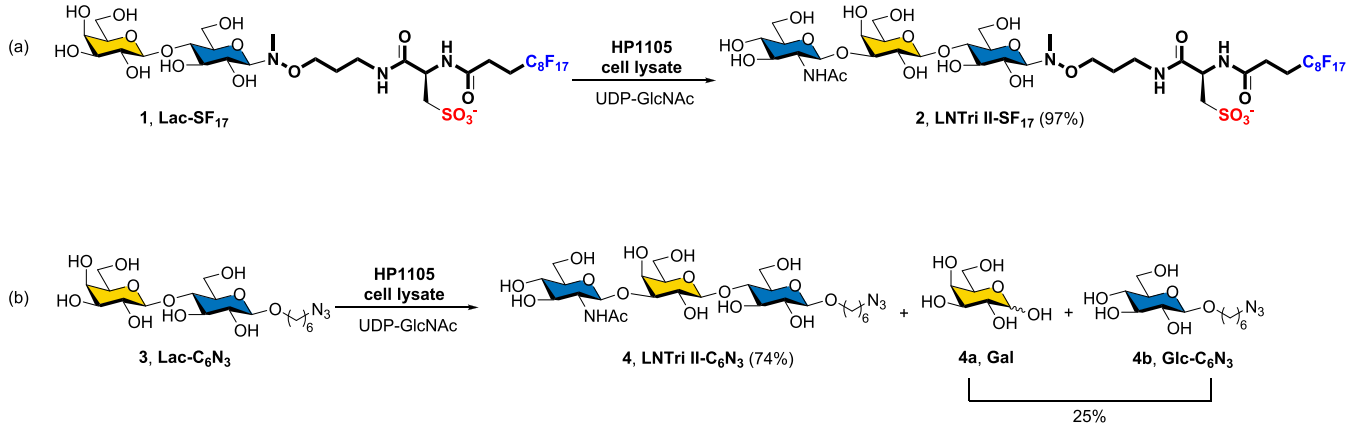
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Scheme 1. Lac-SF_{17} (1) and Lac-C_6N_3 (3) Were Subjected to Enzymatic Elongation Using *E. coli* Cell Lysate Expressing HP1105 (β1,3-*N*-Acetylglucosaminyltransferase), Which Also Contains Endogenous β1,4-Galactosidase (LacZ) Activity[a]



[a]Compound **1** was converted to LNTri II-SF_{17} (**2**) without detectable hydrolysis; while partial Lac-C_6N_3 (**3**) was hydrolyzed under the same condition.

bottleneck remains in purification, which is time-consuming and low-yielding across all methods.

Lin and co-workers previously developed the water-soluble sulfo-fluorous affinity (SOFA) tag, in which installation of a sulfonate moiety improves the aqueous compatibility of the C_8F_{17} fluorous handle and enables rapid purification by fluorous solid-phase extraction after enzymatic glycosylation.[30] The SOFA design also incorporates a photocleavable linker, allowing light-triggered release of amino-functionalized glycans for downstream conjugation, and we observed that the tag can enhance the apparent rates of certain GT reactions. Building on this concept, we subsequently developed the sulfo-fluorous tag (SF_{17}) to enable efficient synthesis and purification while also allowing mild tag removal to access free-reducing HMOs.[31] SF_{17} tag not only enables precise regioselectivity for wild-type fucosyltransferases (FucTs) and sialyltransferases (SiaTs), but also significantly enhances glycosylation efficiency compared to conventional linkers. The unique hydrophobic and amphiphilic nature of SF_{17} facilitates both enzyme–substrate interactions and purification through fluorous solid-phase extraction (FSPE).[32] Moreover, the tag can be cleaved under mildly acidic conditions to afford free reducing-end oligosaccharides for downstream applications, and the recycled SF_{17} tag can be reused in the next conjugation cycle. These properties enabled the synthesis of a diverse panel of complex and highly fucosylated HMOs, including the Lewis x-containing Sia-IFLNH III, trifucosylated TF-*p*LNH II, and TetraF-*p*LNO, many of which are difficult to access by existing methods.

However, despite these advances, achieving robust and scalable synthesis of HMOs remains a significant challenge. Bacterial GTs are increasingly employed in glycan synthesis due to their ease of recombinant expression in *E. coli*. However, large-scale applications require substantial amounts of active enzyme, and many GTs, particularly membrane-bound enzymes, are difficult to purify and often prone to degradation during isolation. As a practical alternative, crude cell lysates are frequently used as enzyme sources to bypass the need for labor-intensive purification. Yet this approach introduces a critical obstacle, as endogenous hydrolases present in the lysates can degrade glycan substrates, thereby compromising reaction efficiency and overall yield. To address this critical bottleneck, we herein present a β-galactosidase -resistant enzymatic strategy using SF_{17}-tagged glycans. The SF_{17} tag not only protects β1,4-galactosyl-capped glycan substrates from hydrolytic degradation in crude lysates but also facilitates selective enzymatic elongation and enables efficient purification via FSPE. This streamlined workflow allows for preparative-scale synthesis of structurally defined, free-reducing end HMOs. By simultaneously addressing the enzymatic vulnerability and purification complexity, our platform offers a practical and scalable solution for producing bioactive glycans, supporting a wide range of applications in glycan–protein interaction studies, structural biology, and glycoengineering.

RESULTS AND DISCUSSION

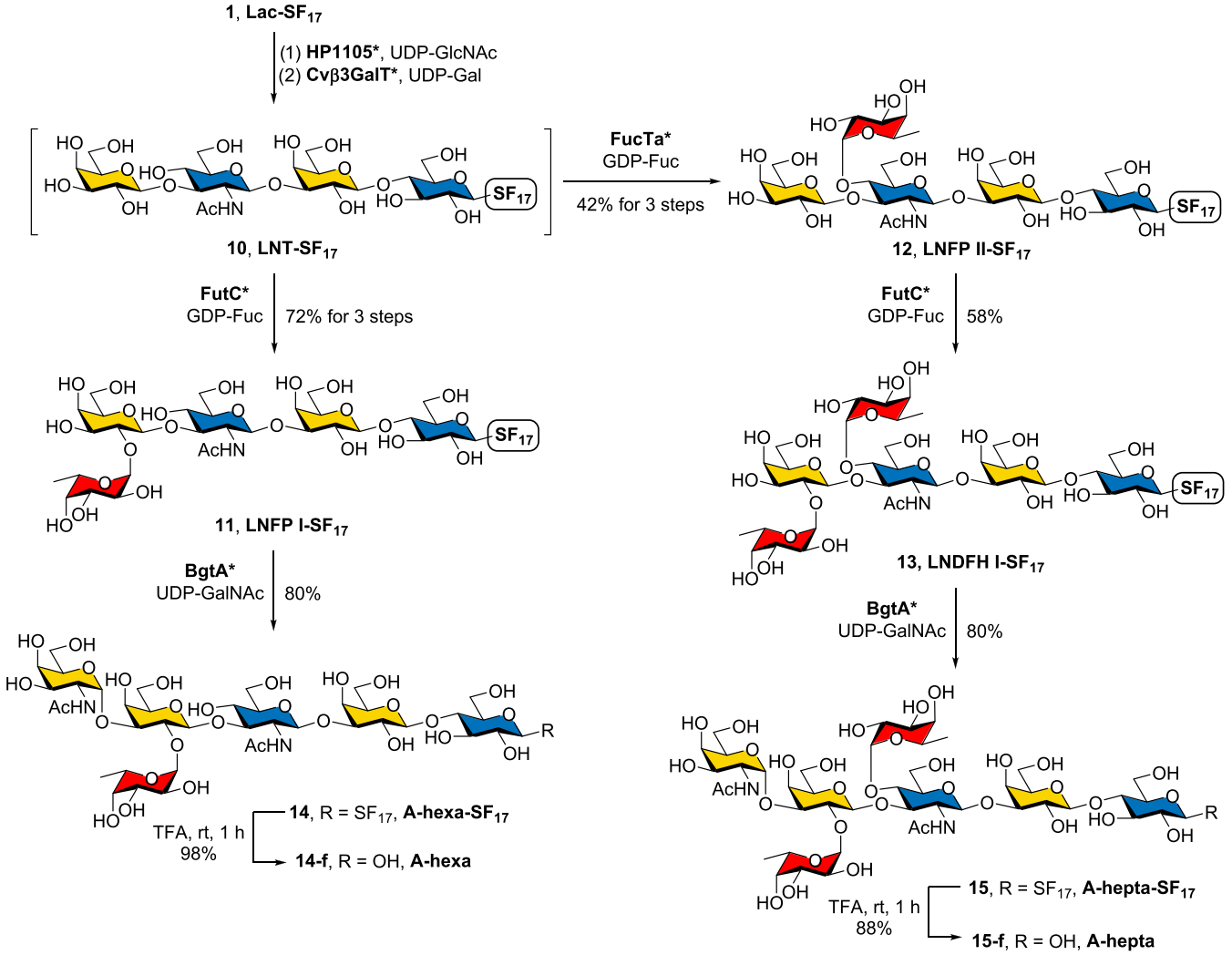
β1,4-Galactosidase Resistance of SF_{17}-Tagged Glycans

To enable efficient preparative-scale synthesis of free-reducing end glycans, a substantial amount of enzymatic activity is required. Thus, simplifying enzyme sourcing is a critical step toward establishing a scalable and practical glycan synthesis platform. In our previous work, HP0826 was used directly in crude *E. coli* lysate for the β1,4-galactosylation of azidohexyl-tagged substrates. Due to the competing activity of endogenous β-galactosidase (LacZ), the reaction had to be carefully quenched immediately after product formation to avoid degradation.[33] To mitigate LacZ-mediated hydrolysis, an intuitive and widely adopted solution is to employ lacZ-deficient (or lacZ-disrupted) *E. coli* hosts for recombinant enzyme expression and lysate preparation. For example, commonly used cloning/expression strains such as JM109 derivatives, as well as commercial T7 expression strains with lacZ disruption (e.g., NEB T7 Express, lacZ::T7 gene1), can substantially reduce β-galactosidase background in Lac(NAc)-based transformations. Nevertheless, in practice, many laboratories, including ours, routinely use BL21(DE3) as a general-purpose host for recombinant GT production. We therefore sought a complementary strategy that could enhance robustness without requiring strain replacement or host re-engineering, and found that installing the SF_{17} tag on Lac(NAc)-containing acceptors substantially reduced their hydrolysis in crude *E. coli* lysate. We initially observed this phenomenon while using crude lysate containing β1,3-*N*-acetylglucosaminyltransferase HP1105,[34] expressed in BL21-

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Scheme 2. Systematic Enzymatic Synthesis of A-Hexa (14-f) and A-Hepta (15-f) from Lac-SF_{17} (1) via Glycosyltransferase-Catalyzed Glycosylation[a]



[a]Reactions were performed using *E. coli* lysates (denoted by *) expressing HP1105, Cvβ3GalT, FutC, FucTa and BgtA, with UDP-sugar donors as indicated. The SF_{17} tag remained stable throughout the multistep synthesis and was removed in the final step using TFA in water to afford free-reducing end A-antigen oligosaccharides **14-f** and **15-f**.

(DE3). In this reaction, Lac-SF_{17} (**1**) was converted to LNTri II-SF_{17} (**2**, 97%) without detectable degradation using HP1105 cell lysate which contains endogenous LacZ, as illustrated in Scheme 1. In contrast, when azidohexyl-tagged lactoside, Lac-C_6N_3 (**3**) was used as the acceptor, the reaction produced not only LNTri II−C_6N_3 (**4**, 74%) but also hydrolyzed byproducts, including galactose (**4a**) and glucoside, Glc-C_6N_3 (**4b**), resulting in an estimated 25% material loss. This issue of the lactoside hydrolysis by LacZ[35] is expected to become more pronounced at preparative scale. To further validate this observation, we tested the susceptibility of lactoside **1** and **3** to purified LacZ. Consistent with our lysate-based results, only Lac-C_6N_3 (**3**) underwent hydrolysis, while Lac-SF_{17} (**1**) remained intact (Figure S1a,b). We then evaluated tetrasaccharide and hexasaccharide substrates containing one or two type-2 LacNAc units to assess their susceptibility to LacZ-mediated hydrolysis. The SF_{17}-tagged glycans, LNnT-SF_{17} (**5**) and *p*LNnH-SF_{17} (**7**), remained intact even after prolonged incubation (Figure S1c). In contrast, the azidohexyl-tagged glycans, LNnT-C_6N_3 (**6**) and *p*LNnH-C_6N_3 (**8**), underwent notable hydrolysis within approximately 1 h (Figure S1d,e). Preincubation of LacZ with free SF_{17} for a defined period, followed by addition of Lac-C_6N_3 (**3**), still resulted in hydrolysis (Figure S1f), providing no evidence for measurable inhibition of LacZ by free SF_{17} under our assay conditions. These results suggest that β1,4-galactose−terminated glycans bearing an SF_{17} tag are more resistant to LacZ activity and suitable for enzymatic elongation in crude lysate systems. While the precise mechanism underlying this resistance remains unclear, we speculate that the sulfonate functionality of SF_{17} may reduce productive interactions with LacZ. Further investigation into this effect may offer deeper insights, but is beyond the scope of the current study.

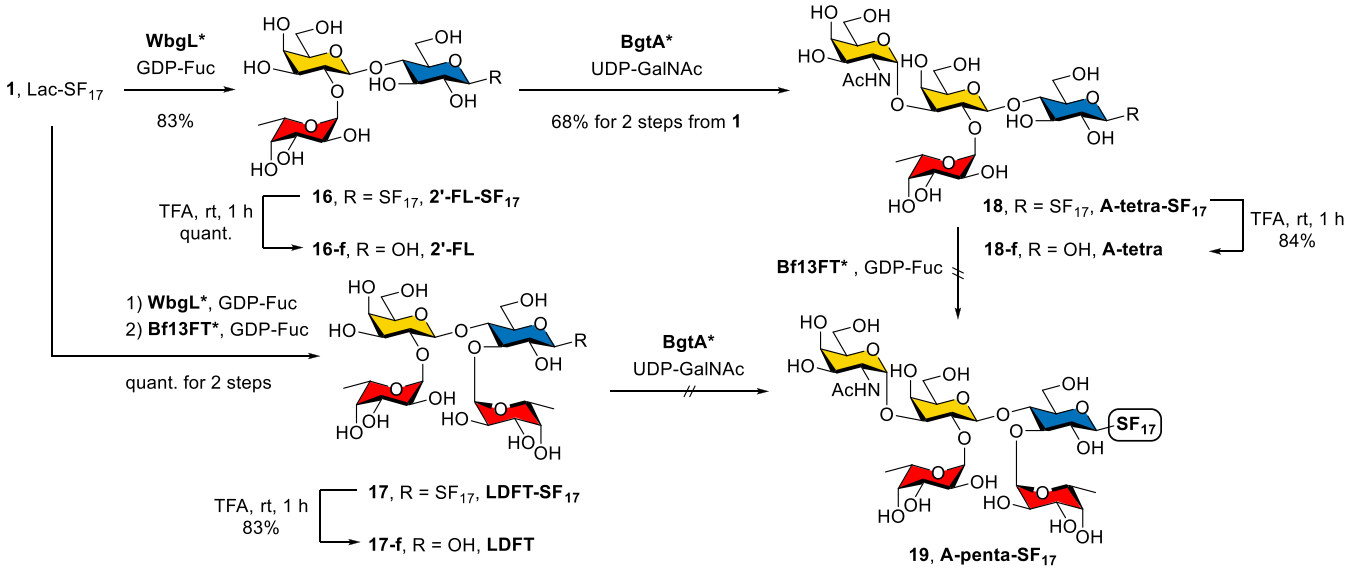
Expanding the Synthetic Scope for Blood Group A Antigen by a Sequential One-Pot Synthesis Approach

Building on the observed resistance of SF_{17}-tagged glycans to hydrolysis, we next explored their compatibility with sequential one-pot enzymatic synthesis. Using *E. coli* lysates expressing HP1105 and Cvβ3GalT (β1,3-galactosyltransferase),[36] we converted Lac-SF_{17} (**1**) into LNT-SF_{17} (**10**) via a two-step,

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Scheme 3. SF_{17}-Assisted Enzymatic Synthesis of Simple HMOs Including 2′-Fucosyllactose (2′-FL, 16-f), Lactodifucotetraose (LDFT, 17-f), and A-Tetra (18-f) using Lac-SF_{17} (1) as the Common Starting Material[a]



[a]Sequential glycosylation reactions were carried out using *E. coli* cell lysates (denoted by *) expressing WbgL, Bf13FT, and BgtA, along with GDP-Fuc and UDP-GalNAc as sugar donors. The SF_{17} tag allowed streamlined synthesis and purification, and was removed in the final step under mild acidic conditions to yield the free-reducing end target glycans.

one-pot protocol (Scheme 2). For HP1105 enzyme purification, each run typically consumes approximately 600 mL of buffer; using crude lysate bypassed this step while maintaining activity and yield. From this intermediate, two synthetic routes were pursued to generate distinct HMO structures. In the first pathway, LNT-SF_{17} (**10**) was fucosylated at the terminal Gal unit by FutC (α1,2-fucosyltransferase from *Helicobacter pylori*)[33] to afford LNFP I-SF_{17} (**11**) in an overall 72% yield across three steps from **1**. In the alternative route, the regioselectivity-enhancing property of the SF_{17} tag enabled selective fucosylation of the GlcNAc residue by FucTa (α1,3/4-fucosyltransferase from *H. pylori*),[31] resulting in LNFP II-SF_{17} (**12**) in 43% overall yield. These results underscore the effectiveness of our strategy in facilitating multistep glycan construction in complex lysate environments, while also minimizing purification steps to improve overall yield.

We further applied this strategy to the synthesis of type A antigen HMOs, as these structures are linked to blood group A activity in the gut and are thought to modulate microbiota composition, gut barrier function, and resistance to enteric pathogens.[37] Histo-blood group antigens (HBGAs) are critical targets in biomedical research, from studying host−pathogen interactions to diagnostics and the design of immunotherapies and cancer vaccines. Synthetic A antigens also serve as useful probes for vaccine development and glycan-mediated cellular recognition.[38,39] Since BgtA (α1,3-*N*-acetylgalactosaminyltransferase from *Helicobacter mustelae*) specifically recognizes the Fucα1 → 2Gal epitope,[40] we used BgtA-containing lysates with UDP-GalNAc to glycosylate glycans **11** and **13**, affording A-hexa-SF_{17} (**14**) and A-hepta-SF_{17} (**15**), each in 80% yield. Importantly, the SF_{17} tag remained stable throughout these transformations, confirming its compatibility with GT-mediated elongation and resistance to endogenous hydrolase activity in *E. coli* lysates. Finally, treatment with 0.1−0.2% trifluoroacetic acid (TFA) in water for 1 h efficiently removed the SF_{17} tag, yielding free-reducing HMOs **14-f** and **15-f** in 98% and 88% yield, respectively.

SF_{17}-Enabled Access to Simple Glycans 2′-FL and LDFT with Streamlined Purification

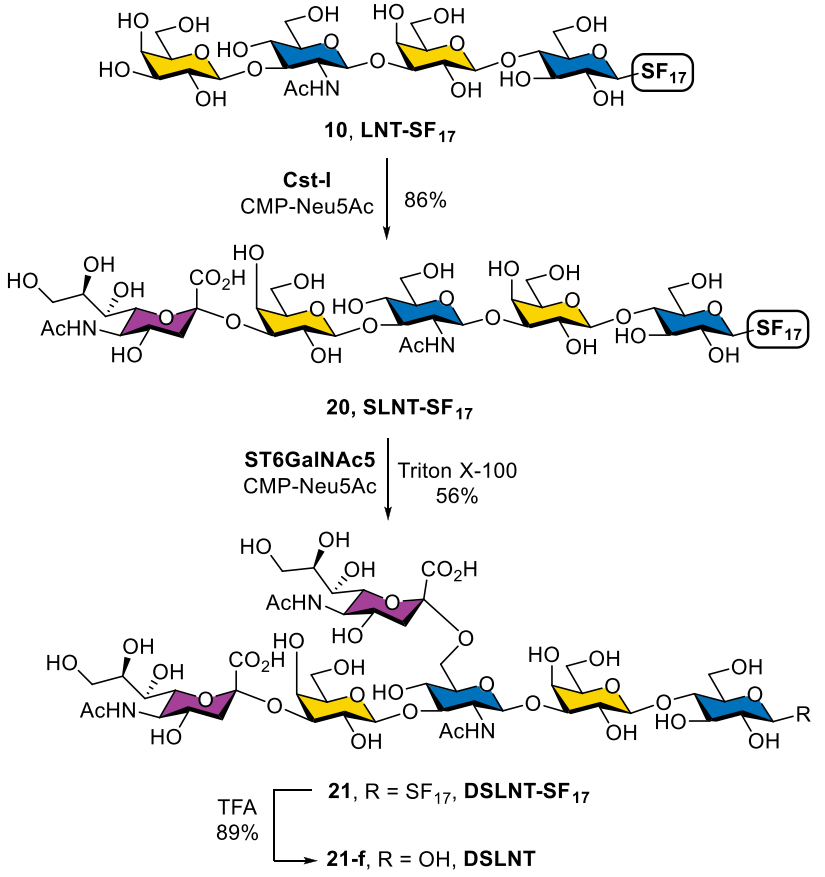
In addition to complex oligosaccharides, we next applied the SF_{17} strategy to the synthesis of simple HMOs, specifically targeting 2′-fucosyllactose (2′-FL, **16-f**) and lactodifucotetraose (LDFT, **17-f**). These bioactive glycans are FDA-approved as food additives and are prominent constituents of human milk, contributing to microbiota development, immune modulation, and protection against pathogens. Their health benefits have also led to growing interest in applications related to gut health, neurodevelopment, and functional food design.[41,42] Fermentation-based production is commonly used for industrial-scale synthesis of these HMOs, while *in vitro* enzymatic synthesis offers superior structural precision.[20] Although FutC has been employed for *in vivo* 2′-FL production in engineered *E. coli*, its limited *in vitro* activity hinders broader synthetic use.[33] As an alternative, we used lysates expressing WbgL (α1,2-fucosyltransferase from *E. coli*),[43] which efficiently catalyzed the conversion of Lac-SF_{17} (**1**) to 2′-FL-SF_{17} (**16**) with improved yield of 83% (Scheme 3). Subsequent fucosylation by Bf13FT (α1,3/4-fucosyltransferase from *Bacteroides fragilis*)[44] afforded LDFT-SF_{17} (**17**) in quantitative yield over two steps. In parallel, based on BgtA's substrate preference for Fucα1 → 2Gal termini, a sequential one-pot reaction using WbgL followed by BgtA cell lysates with GDP-Fuc and UDP-GalNAc yielded A-tetra-SF_{17} (**18**) in 68% overall yield. Attempts to synthesize the A-antigen pentaose (**19**) were unsuccessful using either BgtA on **17** or Bf13FT on **18**.

Due to their low molecular weight and high polarity, simple HMOs such as 2′-FL and LDFT pose considerable purification challenges. Their physicochemical properties closely resemble those of common buffer components (e.g., Tris, phosphate, salts), making conventional methods like ion-exchange, size-exclusion, or HPLC inefficient and labor-intensive. While recent industrial advances (e.g., nanofiltration, electrodialysis,

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Scheme 4. SF_{17}-Assisted Enzymatic Synthesis of DSLNT (21-f)[a]



[a]Sequential sialylation of LNT-SF_{17} (**10**) was carried out using Cst-I, and ST6GalNAc5, with CMP-Neu5Ac (CMP-sialic acid) as the donor. The mono-sialylated SLNT-SF_{17} (**20**) was efficiently converted to DSLNT-SF_{17} (**21**) following detergent-enhanced α2,6-sialylation. Final acidic removal of the SF_{17} tag afforded free-reducing form of DSLNT.

simulated moving bed chromatography) have enabled large-scale purification,[45] such approaches require significant infrastructure and expertise. In contrast, our FSPE-based method offers a rapid, low-cost alternative well-suited for preparative synthesis at the laboratory scale. SF_{17}-tagged glycans were efficiently isolated from crude reaction mixtures without chromatography, and free-reducing HMOs (**16-f** to **18-f**) were obtained in high purity and yield. This strategy overcomes one of the key bottlenecks in simple glycan synthesis and expands access to bioactive free-reducing glycans for downstream applications. While microbial fermentation is highly effective for producing commodity HMOs such as 2′-FL and 3-FL at very large scale, accessing more complex and precisely defined glycans often requires substantial, target-specific pathway engineering and can lead to metabolic burden and product heterogeneity. In contrast, our SF_{17}-enabled *in vitro* workflow provides modular, stepwise enzymatic control and avoids competing cellular pathways, enabling predictable formation of structurally defined products. Together with fluorous-assisted purification, this platform is therefore complementary to fermentation and offers a practical route to complex glycans that remain challenging to obtain microbially.

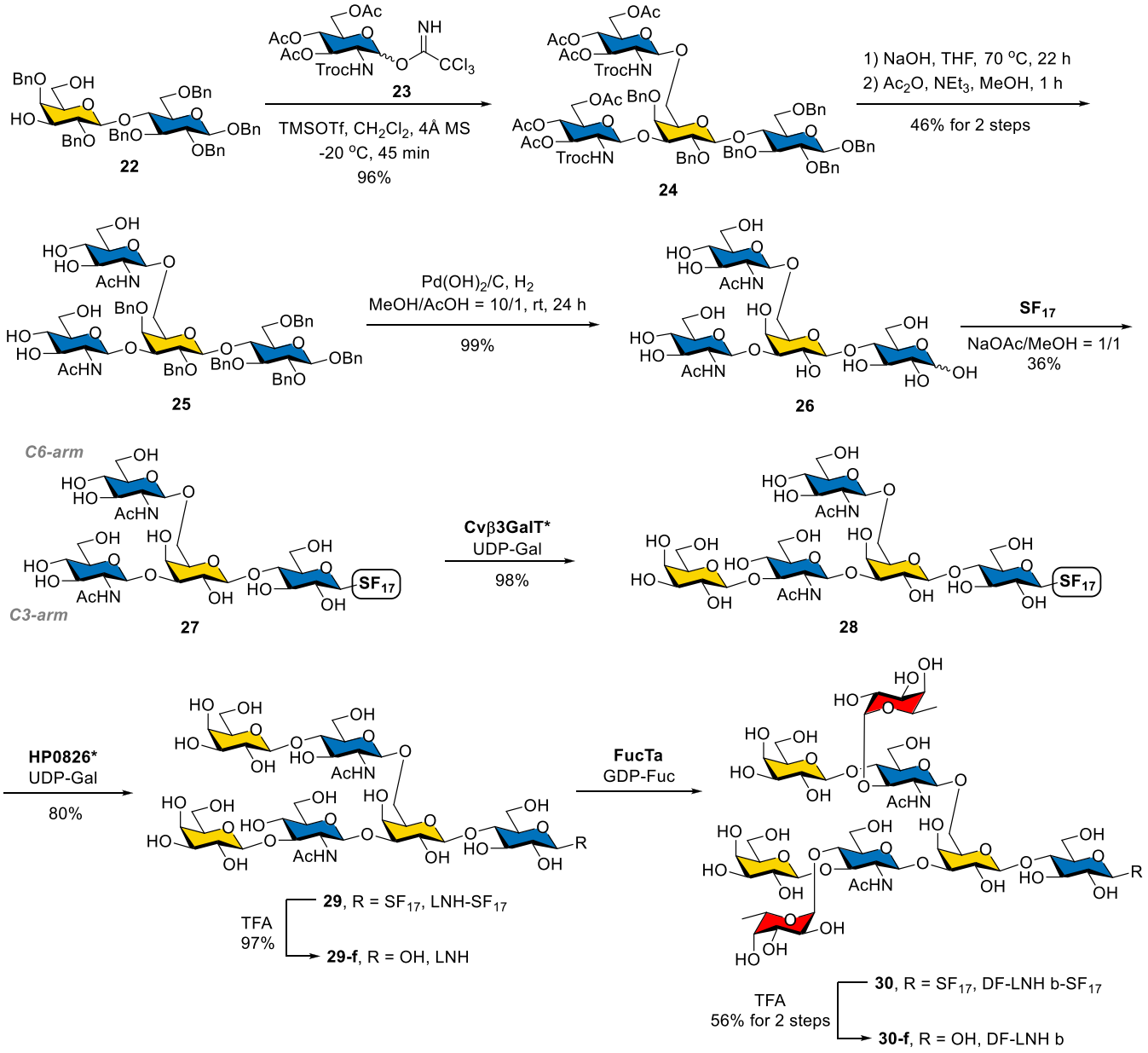
SF_{17}–Assisted Synthesis of DSLNT via Enzymatic Sialylation

We next turned our attention to the synthesis of disialyllacto-*N*-tetraose (DSLNT, **21-f**), an HMO with well-established biological significance. DSLNT has been shown to confer protection against necrotizing enterocolitis (NEC), a severe gastrointestinal condition affecting premature infants.[46−48] In addition to its therapeutic potential, DSLNT is considered a promising biomarker for NEC risk and a candidate for targeted nutritional supplementation. Owing to its clinical relevance, DSLNT has been the subject of several synthetic and analog studies.[49−52] However, while most reported synthetic DSLNT structures feature an anomeric modification, a few studies have successfully obtained the free-reducing form by employing designed substrates followed by hydrogenolysis.[27,53] Complementary enzymatic routes have also been reported to access the internally α2,6-sialylated LacNAc motif relevant to DSLNT. In particular, human ST6GalNAc enzymes, such as ST6GalNAc5 and ST6GalNAc6, can install an α2,6-linked Neu5Ac onto the GlcNAc residue within LacNAc-containing acceptors, although their acceptor preferences differ. Recent studies suggest that an α2,3-sialylated motif in the acceptor can be important for efficient ST6GalNAc5-catalyzed sialylation, whereas this requirement is less stringent for ST6GalNAc6.[50,54] These reports further highlight both the feasibility and the practical challenges of assembling DSLNT and related motifs, motivating streamlined approaches that provide well-defined products while minimizing additional protecting-group or anomeric-handle manipulations. Building on this concept, we applied our SF_{17}-based strategy to synthesize DSLNT in its native free-reducing form. This approach further streamlines

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Scheme 5. SF_{17}-Tagged Strategy for the Synthesis of LNH (29-f) and DF-LNH b (30-f) from a Chemically Prepared Tetrasaccharide Precursor (26)[a]



[a]The intermediate **27** was selectively galactosylated at the C3-arm by Cvβ3GalT to form **28**, followed by C6-arm extension using HP0826 cell lysate (denoted by *) to yield LNH-SF_{17} (**29**). Subsequent di-fucosylation with FucTa afforded DF-LNH b-SF_{17} (**30**), which was deprotected to provide the free-reducing end glycan **30-f**.

access to bioactive HMO structures without requiring additional deprotection steps, and demonstrates the utility of our method for preparing clinically relevant glycans.

In the construction of DSLNT, LNT-SF_{17} (**10**) was first α2,3-sialylated by purified Cst-I from *Campylobacter jejuni*,[55] yielding SLNT-SF_{17} (**20**) in 86% yield. Purified enzyme was used instead of crude lysate because the low activity of Cst-I in lysate required extended reaction times, which led to significant donor hydrolysis. In the subsequent α2,6-sialylation step catalyzed by ST6GalNAc5,[54] however, only trace amounts of product were observed by TLC analysis (see Figure S2 in Supporting Information). Interestingly, the addition of a mild detergent, Triton X-100 (0.1%), markedly improved the reaction outcome, increasing the yield to 56%, likely by reducing substrate aggregation. This modification enabled the efficient synthesis of DSLNT-SF_{17} (**21**), which was subsequently converted to free-reducing DSLNT (**21-f**) in 89% isolated yield (Scheme 4). The ability to obtain native DSLNT may facilitate further studies on NEC-associated glycans.

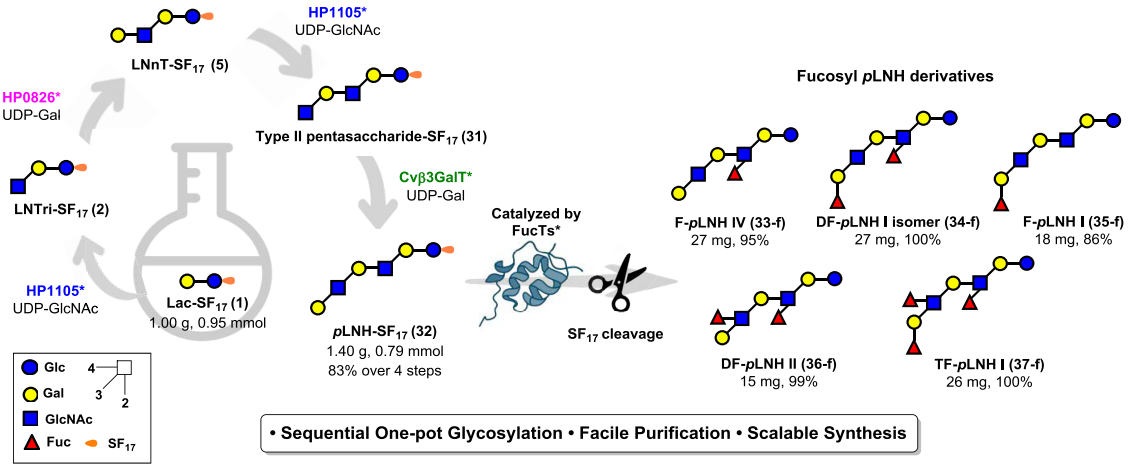
Regioselective Construction of Branched HMOs Using SF_{17}–Tagged Tetrasaccharide Intermediates

To evaluate the applicability of SF_{17} conjugation to larger glycans and its utility in asymmetric HMO branching, we synthesized a core tetrasaccharide (**26**) as a precursor for regioselective enzymatic glycosylation. The construction of tetrasaccharide **26** was achieved through simultaneous installations of the glucosamine residue on **22**, as shown in Scheme 5. Details of the synthesis of **22** can be found in the

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Scheme 6. SF_{17}-Enabled Sequential One-Pot Synthesis of *p*LNH-SF_{17} (32) and Its Fucosylated Analogs at Preparative Scale[a]



[a]Reactions were performed using *E. coli* lysates (denoted by *) expressing HP1105, HP0826, Cvβ3GalT, Bf13FT, FucTa, and FutC. Later acidic cleavage of the SF_{17} tag afforded free-reducing fucosylated *p*LNH derivatives (**33-f**–**37-f**).

Supporting Information (Schemes S2 and S3). Glycosylation of the C3′- and C6′-*O*-unprotected acceptor **22** with glycosyl donor **23** proceeded at −20 °C in the presence of catalytic trimethylsilyl trifluoromethanesulfonate (TMSOTf), and the desired tertrasaccharide **24** was obtained in 94% yield with complete β-stereoselectivity. The 2,2,2-trichloroethoxycarbonyl (Troc) group of **24** was converted to an acetyl group by executing a two-step protecting group manipulation, which resulted in **25** (46% yield over 2 steps). Accordingly, concomitant deprotection of the Troc and ester groups of **24** was accomplished by using sodium hydroxide in THF. The crude product was then desalted using a C18 solid-phase extraction (SPE) cartridge, which was followed by treatment with acetic anhydride and triethylamine (NEt_3) to convert amine to acetamide. Benzyl ethers subsequently underwent hydrogenolysis by palladium hydroxide on carbon using a hydrogen balloon. After the desalting steps by the combined use C18 SPE cartridge and HW-40F size-exclusion chromatography, the key tetrasaccharide precursor **26** for SF_{17} conjugation was obtained in 99% yield. Conjugation of SF_{17} to tetrasaccharide **26** was achieved under acidic conditions (0.1 M NaOAc (pH 4.5)/MeOH = 1/1(v/v)), producing the SF_{17}-tagged intermediate **27** in 36% yield. The relatively low yield was attributed to the reduced conformational flexibility of the large oligosaccharide, which may hinder the ring-opening/ring-closing process required for efficient oxime formation.[56] Despite incomplete conversion, tetrasaccharide **26** was fully recyclable and could be reused in the next round of conjugation.

We then explored the regioselective enzymatic glycosylation of **27** to construct asymmetrically branched HMOs (Scheme 5). Cvβ3GalT[26] catalyzed exclusive β1,3-galactosylation on the C3-arm of **27**, affording pentasaccharide **28** in 98% yield. While our manuscript was under preparation, a similar regioselectivity was independently reported by the Cao group.[57] Subsequently, the C6-arm GlcNAc residue of **28** was elongated via HP0826-catalyzed β1,4-galactosylation, yielding LNH-SF_{17} (**29**) in 80% isolated yield. The SF_{17} tag was then removed to give the free-reducing form LNH (**29-f**) in 97% yield. We also demonstrated that SF_{17}-tagged tetrasaccharide substrates performed significantly better than their free-reducing counterparts in crude lysate systems, enabling efficient regioselective β1,3-galactosylation and downstream glycan elongation. (see Scheme S1 in the Supporting Information). Further difucosylation of **29** by FucTa afforded DF-LNH b-SF_{17} (**30**), which was converted to DF-LNH b (**30-f**) in 56% yield over two steps after SF_{17} tag removal. Purified FucTa was used for this step due to the poor performance of the lysate. DF-LNH b is a major structural component of N3 antigens, which contain both Lewis a (Le^a) and Lewis x (Le^x) motifs. Notably, Le^x expression is elevated in early stage gastric cancer and has been linked to gastrointestinal tumor progression.[24] Our approach provides access to this biologically relevant structure, potentially enabling studies on tumor-associated glycoepitopes and biomarker development.

Preparative-Scale Synthesis of *para*-Lacto-*N*-Hexaose and Its Fucosylated Derivatives Using the SF_{17} Strategy

To validate scalability, we applied the SF_{17} strategy to the gram-scale synthesis of *para*-lacto-*N*-hexaose-SF_{17} (*p*LNH-SF_{17}, **32**). Starting from Lac-SF_{17} (**1**), a four-step sequential one-pot glycosylation was executed with *E. coli* lysates expressing HP1105, HP0826, HP1105, and Cvβ3GalT, respectively (Schemes 6 and S4). After each step, the GT was heat-inactivated prior to addition of the next enzyme and the corresponding sugar nucleotide. Without intermediate purification of **2**, **5**, and **31**, this sequence enabled a streamlined route. From 1.0 g of starting material **1**, we successfully isolated 1.40 g of **32** in 5 days, corresponding to 83% overall yield over 4 steps, thereby demonstrating the preparative-scale potential. Greenness metrics were evaluated on a route level (aggregated across the four sequential one-pot steps). The process mass intensity (PMI) was 1540 including water and 592 excluding water, with the following contributions: substrates/reagents 15, organic solvents 577, and water 948. System boundaries: net fresh inputs, organic solvents counted as net makeup after recovery, no byproduct credit (see Tables S1 and S2 in Supporting Information).

To further showcase the versatility of this platform, we extended the strategy to generate a series of structurally diverse fucosylated *p*LNH derivatives **33**–**37**. By employing different α1,2- and α1,3/4-FucTs, including WbgL and FucTa, fucosylation was selectively introduced at specific positions on the *p*LNH scaffold. The resulting SF_{17}-tagged glycans were

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then deprotected to yield the corresponding free-reducing end forms **33-f** to **37-f** (see details in Supporting Information). All glycan structures were fully characterized by NMR spectroscopy to confirm their identities (Tables S3−S6). In our previous work,[31] these compounds were synthesized only in small quantities, with structural analysis primarily dependent on mass spectrometry. In contrast, the present study demonstrates not only the feasibility of producing these glycans on a preparative scale, but also enables access to structurally defined glycans, facilitating downstream applications in glycan−protein interaction studies and structural biology. A comparison of scales and yields between the two studies is provided in Table S7.

CONCLUSIONS

In conclusion, we have developed a robust and scalable glycan synthesis strategy using the sulfo-fluorous affinity tag SF_{17}. Conjugation of SF_{17} to the reducing end of glycans bearing terminal β1,4-galactose residues provides effective protection against endogenous β-galactosidase activity in crude *E. coli* lysates. Coupling SF_{17} with FSPE reduces buffer and solvent consumption during purification, lowers unit operation, and improves overall process efficiency. This protection allows the tagged glycans to remain intact and function as stable acceptors or products during subsequent GT-catalyzed reactions. In addition, the SF_{17} tag promotes regioselective glycan elongation and enables efficient purification via FSPE. Together, these features establish SF_{17} as a unified solution to multiple challenges in preparative-scale, free-reducing glycan synthesis. The generality and utility of this platform were demonstrated across a broad range of HMOs, including simple structures such as 2′-FL (**16-f**) and LDFT (**17-f**), complex frameworks like DSLNT (**21-f**) and DF-LNH b (**30-f**), A-antigen-type HMOs including A-tetra (**18-f**), A-hexa (**14-f**), and A-hepta (**15-f**), and fucosylated derivatives (**33-f**−**37-f**) based on *para*-lacto-*N*-hexaose (*p*LNH). Notably, the successful gram-scale synthesis of *p*LNH-SF_{17} (**32**) and its fucosylated derivatives highlights the method's preparative viability. In contrast to our earlier work, which was constrained by limited reaction scale and purification inefficiencies, the current strategy affords access to structurally defined glycans in quantities sufficient for biochemical analysis, structural biology, and *in vitro* evaluation. Overall, this SF_{17}-enabled workflow provides a powerful and generalizable platform for functional glycomics. Its compatibility with crude enzyme systems, streamlined purification, and scalability make it especially attractive for glycan−protein interaction studies, therapeutic glycan development, and industrial applications.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.5c10643.

Detailed experimental procedures, enzyme purification, enzyme kinetics plots, and ^{1}H NMR and ^{13}C NMR spectra illustrating chemical structures for enzymatic products (PDF)

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Author Contributions

M.-T.L. performed the chemical and enzymatic synthesis, as well as protein expression. C.-E.Y. and Y.-T.H. prepared the glycan acceptors and sugar nucleotide donors for enzymatic reactions. M.-T.L. and C.-C.Y. designed the overall synthetic strategy, analyzed the data, and wrote the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We acknowledge Chu-Lan Peng, I-Ying Lai, Dr. Chu-Ya Wu, and Dr. Hsin-Ru Wu at the Instrumentation Center of National Tsing Hua University for their support with the NMR and ESI-MS analyses, as well as Ping-Yu Lin at the Institute of Chemistry, Academia Sinica, for assistance with ESI-MS experiments. We thank Chang-Lun Lo at National Chung Cheng University for assistance with the NMR experiments. We thank Dr. Hsing-Kai Tseng for providing ST6GalNAc5. This work was financially supported by National Tsing Hua University, National Chung Cheng University, and the National Science and Technology Council (NSTC) of Taiwan (NSTC 112-2113-M-007-026-MY3 and NSTC 110-2628-M-007-007-MY3).

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